A CASE-STUDY BASED REPORT

ON

<Early-Stage Diabetes Risk Prediction Using Random Forest and XGBoost Classifiers >

submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

in

A.Y.: 2026-27

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Abstract

Diabetes is a major health condition that requires early identification and appropriate medical attention. Machine learning techniques can help analyze patient-related characteristics and symptoms to identify patterns associated with diabetes risk. This project, titled “Early-Stage Diabetes Risk Prediction Using Random Forest and XGBoost Classifiers,” aims to develop a machine learning-based system that predicts whether a given patient record belongs to the Positive or Negative class based on demographic information and symptoms.

The project uses the UCI Early Stage Diabetes Risk Prediction Dataset, which originally contains 520 records, 16 input features, and one target variable. Exploratory Data Analysis (EDA) was performed to understand the dataset structure, examine missing values, identify duplicate records, study feature distributions, and analyze correlations. The dataset contained no missing values and 269 duplicate records. After removing duplicates, 251 unique records remained. Categorical variables were converted into numerical values using binary mapping, and the dataset was divided into training and testing sets using an 80:20 stratified split.

Two ensemble machine learning algorithms, Random Forest and XGBoost, were implemented and evaluated using accuracy, precision, recall, F1-score, and confusion matrices. On the test set, Random Forest achieved an accuracy of 92.16% and an F1-score of 93.94%, while XGBoost achieved an accuracy of 88.24%. Five-fold cross-validation was also performed to assess model performance across different data splits.

Finally, the trained models were integrated into a Flask-based web application that accepts patient information and displays predictions from both classifiers. The system is intended to support preliminary risk screening and is not a substitute for professional medical diagnosis.

Introduction

2.1 Background

Diabetes is a chronic health condition that affects the body's ability to regulate blood glucose levels. Early identification of potential diabetes risk is important because timely medical evaluation and appropriate treatment can help reduce the risk of complications. However, symptoms may vary among individuals, and some symptoms may not be recognized during the early stages.

Machine learning has become a useful approach for analyzing healthcare-related datasets and identifying patterns among multiple patient characteristics. By examining demographic information and symptoms, machine learning classification algorithms can learn relationships between input features and known class labels.

This project, titled “Early-Stage Diabetes Risk Prediction Using Random Forest and XGBoost Classifiers,” applies two ensemble learning algorithms to classify patient records into Positive or Negative categories. The system also includes a web-based interface through which users can enter patient information and obtain predictions from the trained models.

2.2 Motivation for the Project

The primary motivation for this project is to explore how machine learning can support the preliminary identification of potential diabetes risk using patient-related information and symptoms. Analyzing several symptoms together may help identify patterns that are not easily understood by considering individual symptoms separately.

Another motivation is to compare the performance of two ensemble classification algorithms, Random Forest and XGBoost, on the same dataset. Evaluating both models using multiple performance metrics provides a better understanding of their classification performance.

The project also aims to make the prediction process accessible through a simple web interface. By combining data analysis, machine learning, and web development, the project demonstrates how a trained model can be integrated into an interactive application. The resulting system is intended for educational and preliminary screening-support purposes rather than independent medical diagnosis.

2.3 Overview of the Work Carried Out

The project was completed through a sequence of data analysis, preprocessing, model development, evaluation, and web application integration steps.

First, the UCI Early Stage Diabetes Risk Prediction Dataset was loaded and examined using Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn. Exploratory

Data Analysis was performed to understand the dataset structure, inspect missing values, identify duplicate records, examine the target-class distribution, study age and gender distributions, analyze symptoms, and visualize feature correlations.

During preprocessing, duplicate records were removed and categorical variables were converted into numerical representations using binary mapping. The input features were separated from the target variable, and the dataset was divided into training and testing subsets using an 80:20 stratified split.

Next, two classification models were implemented: Random Forest and XGBoost. Their performance was evaluated using accuracy, precision, recall, F1-score, and confusion matrices. Five-fold cross-validation was also used to examine performance across multiple data splits.

Finally, the trained models were saved and integrated into a Flask-based web application. The application accepts patient information and symptoms, processes the inputs, and displays the predictions generated by both models.

2.4 Scope and Responsible-Use Statement

The scope of this project is limited to developing and evaluating machine learning classifiers using the selected early-stage diabetes dataset. The system uses 16 input features, including age, gender, and health-related symptoms, to predict the Positive or Negative class. It compares Random Forest and XGBoost and demonstrates their use through a web-based interface.

The project focuses on data preprocessing, exploratory analysis, classification, model evaluation, and application integration. Future improvements may include using larger and more diverse datasets, incorporating additional clinical parameters, performing more extensive hyperparameter tuning, and evaluating the models on independent datasets.

The system has important limitations. Its predictions depend on the quality, size, and characteristics of the dataset used for training. The performance obtained during testing does not guarantee the same results for every population or individual. In addition, a model prediction does not establish the presence or absence of diabetes.

Responsible-use statement: This application is intended for educational purposes and preliminary risk-screening support only. It is not a medical diagnostic tool and must not replace laboratory testing, professional medical advice, or evaluation by a qualified healthcare professional.

Problem Statement

3.1 Case-Study Scenario

Diabetes is a significant health concern, and identifying individuals who may be at risk at an early stage can support timely medical evaluation. Individuals may experience different symptoms, such as excessive urination, excessive thirst, sudden weight loss, weakness, increased hunger, and blurred vision. These symptoms may occur in different combinations, making it useful to analyze multiple patient characteristics together.

In this case study, the UCI Early Stage Diabetes Risk Prediction Dataset is used to investigate whether machine learning algorithms can identify patterns associated with the Positive and Negative classes. The original dataset contains 520 records, 16 input features, and one target variable. After removing duplicate records, 251 unique records remain for model development and evaluation.

Two ensemble classification algorithms, Random Forest and XGBoost, are applied to the cleaned dataset. Their performance is compared using standard classification metrics. The trained models are subsequently integrated into a Flask-based web application that accepts patient information and symptoms and displays the resulting predictions.

This case study demonstrates the application of data analysis and machine learning to preliminary diabetes risk screening while recognizing that predictions must not be treated as confirmed medical diagnoses.

3.2 Central Case-Study Question

The central question guiding this study is:

“How effectively can Random Forest and XGBoost classifiers predict the Positive or Negative class associated with early-stage diabetes using patient demographic information and symptoms, and how can these models be integrated into a web-based prediction system?”

The study addresses this question by examining the dataset, preprocessing the available features, training both classification algorithms, evaluating their performance, and comparing the resulting metrics. It also demonstrates how the trained models can be made accessible through a simple web application.

3.3 Formal Problem Definition

The problem is formulated as a supervised binary classification task. The dataset contains patient-related input features and a known target label, which allows machine learning algorithms to learn patterns from labeled examples.

Let the input feature vector be represented as:

$X = (x_1, x_2, x_3, ..., x_{16})$

where the 16 features include age, gender, and health-related symptoms. The target variable is represented by Y, which has two possible classes:

- **Positive (Y = 1):** The record belongs to the Positive class in the dataset.
- **Negative (Y = 0):** The record belongs to the Negative class in the dataset.

The objective is to develop a classification function that maps the input feature vector to one of these two classes:

$Y = f(X)$

Two classification models, Random Forest and XGBoost, are trained using the processed dataset. Their predictions are evaluated on a separate test set using accuracy, precision, recall, F1-score, and confusion matrices.

The problem also includes integrating the trained classifiers into a web application so that users can provide the required input features and obtain model-generated predictions. The output represents a prediction based on learned data patterns and does not constitute a clinical diagnosis.

3.4 Objectives that Guide the Study

The major objectives of this study are:

1. **Dataset Understanding:** To examine the UCI Early Stage Diabetes Risk Prediction Dataset and understand its structure, features, and target variable.
2. **Exploratory Data Analysis:** To investigate missing values, duplicate records, class distribution, age distribution, gender distribution, symptom patterns, and feature correlations.
3. **Data Preprocessing:** To remove duplicate records, encode categorical values into numerical representations, and prepare the dataset for model training.
4. **Model Development:** To implement Random Forest and XGBoost classifiers for binary classification of the dataset.
5. **Model Evaluation:** To assess model performance using accuracy, precision, recall, F1-score, and confusion matrices.
6. **Performance Comparison:** To compare the classification results of Random Forest and XGBoost and identify which model performs better on the selected evaluation data.
7. **Cross-Validation:** To use five-fold cross-validation to examine model performance across multiple data splits.
8. **Web Application Integration:** To integrate the trained models into a Flask-based web interface that accepts patient information and displays predictions.
9. **Responsible Use:** To present the system as an educational and preliminary screening-support application rather than a replacement for professional medical evaluation.

3.5 Assumptions and Limitations

Assumptions

The study is based on the following assumptions:

1. The supplied dataset contains meaningful information for investigating patterns associated with the target classes.
2. The target variable, `class`, correctly represents the Positive and Negative labels recorded in the dataset.
3. The input features supplied to the application use the same definitions and numerical mappings as those used during model training.
4. The training and testing datasets are prepared consistently, and the test set is kept separate from model fitting.
5. The predictions generated by the classifiers represent learned patterns in the dataset and are not interpreted as definitive medical conclusions.

Limitations

The study has the following limitations:

1. **Limited Dataset Size:** The original dataset contains 520 records, but only 251 unique records remain after duplicate removal. This limits the amount of unique data available for training and evaluation.
2. **Dataset Representativeness:** The dataset may not represent every age group, geographic region, or population. Consequently, the model's performance may differ when applied to other populations.
3. **Restricted Input Features:** The system relies on age, gender, and the symptoms included in the dataset. It does not incorporate every possible clinical measurement, laboratory result, medical history, or lifestyle factor.
4. **Model Performance:** The reported evaluation metrics are specific to the dataset and evaluation procedure used in this study. They do not guarantee equivalent performance on new or independent datasets.
5. **Potential Classification Errors:** The models may produce false-positive or false-negative predictions. In particular, a Negative prediction does not rule out the possibility of diabetes.
6. **Limited Model Selection:** The project compares Random Forest and XGBoost. Other classification algorithms and more extensive hyperparameter tuning could be investigated in future work.
7. **Clinical Limitation:** The system has not been established as a clinically validated diagnostic tool. Its predictions should not be used independently to make medical decisions.

Despite these limitations, the study provides a practical demonstration of data preprocessing, ensemble learning, model evaluation, and web application integration for preliminary diabetes risk prediction.

DATASET AND EXPLORATORY DATA ANALYSIS

4.1 About the Dataset

The dataset used in this project is the UCI Early Stage Diabetes Risk Prediction Dataset. It contains patient-related demographic information and symptoms that can be used to study patterns associated with early-stage diabetes classification.

The original dataset consists of 520 records and 17 columns. Among these columns, 16 are input features and one is the target variable, named class. The input features include age, gender, and 14 health-related symptoms or characteristics. The target variable contains two classes: Positive and Negative.

Dataset Characteristics

Property	Description
Dataset name	UCI Early Stage Diabetes Risk Prediction Dataset
Original number of records	520
Total number of columns	17
Input features	16
Target variable	class
Target classes	Positive and Negative
Numerical input feature	Age
Categorical input features	15
Missing values	0
Duplicate records identified	269
Unique records after duplicate removal	251
Positive records after cleaning	173
Negative records after cleaning	78

Input Features

The dataset contains the following 16 input features:

1. Age
2. Gender
3. Polyuria
4. Polydipsia
5. Sudden weight loss
6. Weakness
7. Polyphagia
8. Genital thrush
9. Visual blurring
10. Itching
11. Irritability
12. Delayed healing
13. Partial paresis
14. Muscle stiffness
15. Alopecia
16. Obesity

The class column is the target variable that the machine learning models aim to predict. These features allow the classifiers to learn relationships between patient characteristics and the recorded target class.

4.2 Purpose of EDA

Exploratory Data Analysis (EDA) is the process of examining and visualizing a dataset to understand its structure, quality, distributions, and relationships before applying machine learning algorithms.

In this project, EDA was performed to understand the characteristics of the diabetes dataset and identify any data-quality issues that could affect model development.

The main purposes of EDA were:

1. To understand the number of records, columns, and data types in the dataset.
2. To identify missing values and determine whether any imputation was necessary.
3. To detect duplicate records and assess the effect of repeated observations.
4. To examine the distribution of the Positive and Negative target classes.
5. To understand the distribution of patient ages.
6. To analyze gender distribution within the dataset.
7. To investigate the distribution of individual symptoms and health characteristics.
8. To visualize correlations between numerically represented features.
9. To support appropriate preprocessing decisions before training the classifiers.

EDA helped establish a clear understanding of the dataset and guided the preparation of the data for Random Forest and XGBoost.

4.3 Preparation Steps Actually Performed

The dataset was prepared through a sequence of inspection, cleaning, and transformation steps before model training.

4.3.1 Loading the Dataset

The dataset was loaded into a Pandas DataFrame using Python. Pandas was used because it provides convenient functions for reading, inspecting, and manipulating tabular data.

The initial inspection included checking the dataset's dimensions, column names, data types, and sample records.

4.3.2 Checking Missing Values

Missing values were examined using Pandas functions. The analysis showed that the dataset contained zero missing values.

Therefore, no missing-value imputation or deletion was required.

4.3.3 Identifying and Removing Duplicate Records

Duplicate records were identified using the duplicate-checking functionality available in Pandas. The original dataset contained 269 duplicate records.

These duplicate records were removed before model training to reduce repetition and lower the risk of overly optimistic evaluation caused by repeated observations appearing in both training and testing data.

After duplicate removal, 251 unique records remained, consisting of 173 Positive records and 78 Negative records.

4.3.4 Identifying Numerical and Categorical Features

The dataset contained one numerical input feature, Age, and 15 categorical input features. The categorical inputs included Gender and the health-related symptoms represented by Yes/No values.

The target variable, class, was also categorical before encoding.

Understanding these feature types helped determine the appropriate encoding method for preparing the dataset for machine learning.

4.3.5 Encoding Categorical Variables

The categorical values were converted into numerical representations using direct binary mapping.

The mapping used was:

Original Value	Encoded Value
Yes	1
No	0
Male	1
Female	0
Positive	1
Negative	0

This transformation was performed using a dictionary-based replacement operation in Pandas. It is a form of binary encoding and is not one-hot encoding.

Encoding enabled the categorical information to be represented numerically for use by the classifiers.

4.3.6 Separating Input Features and Target

After encoding, the dataset was divided into two components:

- **Input features (X):** The 16 patient-related features used to make predictions.
- **Target variable (y):** The class column containing the Positive or Negative label.

This separation ensured that the models learned from the input features and were evaluated against the correct target labels.

4.3.7 Train-Test Split

The cleaned and encoded dataset was divided into training and testing subsets using an 80:20 split. A stratified split was used to preserve approximately the same target-class proportions in both subsets.

The random_state=42 parameter was used to make the split reproducible.

The resulting dataset contained 200 training records and 51 testing records. The training subset was used to fit the models, while the testing subset was reserved for evaluating their performance.

4.4 EDA Exploration – Implementation

EDA was implemented using Python libraries, including Pandas, NumPy, Matplotlib, and Seaborn. These libraries supported data inspection, numerical analysis, and visualization.

4.4.1 Dataset Structure Analysis

The shape, column names, data types, and sample records were examined to understand the dataset's structure.

Functions such as df.shape, df.columns, df.info(), and df.head() were used for this purpose.

The analysis confirmed that the original dataset contained 520 records and 17 columns.

4.4.2 Missing-Value Analysis

Missing values were checked using the isnull() and sum() functions. The resulting output showed zero missing values across the dataset.

This indicated that missing-value treatment was not necessary before preprocessing.

4.4.3 Duplicate Analysis

Duplicate records were examined using the duplicated() function. A total of 269 duplicate records were identified.

After duplicate removal, the cleaned dataset contained 251 unique records. This step was important for preparing a less repetitive dataset for model development and evaluation.

4.4.4 Target-Class Distribution

The distribution of the target variable was examined using class counts and graphical visualization.

Before duplicate removal, the original dataset contained 320 Positive records and 200 Negative records. After removing duplicates, the class distribution became 173 Positive records and 78 Negative records.

This analysis helped reveal the distribution of the target classes and showed that the cleaned dataset contained more Positive records than Negative records.

4.4.5 Age Distribution

A histogram was used to examine the distribution of patient ages. This visualization helped identify the age ranges represented in the dataset and understand how frequently different ages occurred.

Age distribution analysis provides useful context for interpreting the demographic characteristics of the available observations.

4.4.6 Gender Distribution

Gender distribution was examined using categorical counts and visualization. This analysis helped identify the representation of male and female records in the dataset.

It provided a descriptive overview of the gender composition of the available data.

4.4.7 Symptom Distribution

The distributions of symptoms and health characteristics were examined using categorical plots. The analysis included features such as Polyuria, Polydipsia, sudden weight loss, weakness, Polyphagia, visual blurring, itching, irritability, delayed healing, partial paresis, muscle stiffness, Alopecia, Obesity, and Genital thrush.

These visualizations helped compare the recorded Yes/No responses across symptoms and understand the characteristics of the dataset.

4.4.8 Correlation Analysis

A correlation matrix and heatmap were generated to visualize relationships between the numerically represented features.

The heatmap provided a graphical summary of the direction and strength of pairwise linear associations. Since categorical variables were encoded numerically, the resulting correlations should be interpreted with care.

Correlation analysis can help identify relationships in the dataset, but correlation does not establish causation. The heatmap was used as an exploratory tool rather than as proof that a particular symptom causes diabetes.

4.5 Summary of EDA Findings

The EDA provided an overview of the dataset's quality, composition, and feature distributions before model development.

The main findings were:

1. The original dataset contained 520 records and 17 columns, including 16 input features and one target variable.
2. No missing values were found, so no missing-value imputation was necessary.
3. A total of 269 duplicate records were identified and removed.
4. The cleaned dataset contained 251 unique records.
5. The cleaned target distribution consisted of 173 Positive records and 78 Negative records.

6. Age was the only numerical input feature, while the remaining 15 input features were categorical.
7. The target variable was converted from Positive/Negative labels into numerical values for model training.
8. Age, gender, and symptom distributions were explored through graphical visualizations.
9. A correlation heatmap was used to examine associations between the encoded features.
10. The cleaned and encoded dataset was divided into training and testing subsets using an 80:20 stratified split.

Overall, EDA helped identify duplicate records, confirm the absence of missing values, understand the target-class distribution, and prepare the dataset for classification. The findings provided the foundation for implementing and evaluating Random Forest and XGBoost classifiers.

ALGORITHM SELECTION AND IMPLEMENTATION

5.1 Introduction

The main objective of this project is to develop a machine learning system that classifies patient records into Positive and Negative classes using demographic information and symptoms. Since the target variable contains two possible classes, the problem is formulated as a binary classification task.

Two ensemble learning algorithms, Random Forest and XGBoost, were selected and implemented. Both algorithms combine multiple decision trees but use different approaches to build the final classifier. Their performance was compared to identify which model performed better on the selected dataset and test split.

5.2 Algorithm Selection

5.2.1 Random Forest Classifier

Random Forest is an ensemble machine learning algorithm that combines the predictions of multiple decision trees. Each tree is trained using randomly selected training observations and features. For classification, the final prediction is generally determined through majority voting among the individual trees.

Random Forest was selected because it can model nonlinear relationships, handle numerical and encoded categorical inputs, and reduce the dependence on an individual decision tree.

Implementation parameters:

- Algorithm: `RandomForestClassifier`
- Number of trees (`n_estimators`): 100
- Random state: 42

The model was trained using the training dataset and evaluated using the test dataset.

5.2.2 XGBoost Classifier

XGBoost, or Extreme Gradient Boosting, is an ensemble learning algorithm based on gradient boosting. It builds decision trees sequentially, with each new tree contributing to the improvement of the existing model by reducing the loss function.

XGBoost was selected to compare the performance of a boosting-based approach with the bagging-based Random Forest algorithm. It can capture complex patterns and relationships within the input features.

Implementation parameters:

- Algorithm: `XGBClassifier`
- Number of estimators (`n_estimators`): 100
- Maximum tree depth (`max_depth`): 3
- Learning rate: 0.1
- Evaluation metric: `logloss`
- Random state: 42

The model was trained using the same training dataset as Random Forest to support a fair comparison.

5.3 Difference Between Random Forest and XGBoost

Random Forest and XGBoost are both ensemble methods, but their training strategies differ.

Feature	Random Forest	XGBoost
Ensemble approach	Bagging	Gradient boosting
Tree construction	Trees are built independently	Trees are built sequentially
Main objective	Combine predictions and reduce variance	Improve predictions by reducing errors and loss
Prediction combination	Majority voting for classification	Combined weighted contributions of trees
Main parameter in this project	100 trees	100 estimators, depth 3, learning rate 0.1

The comparison helps demonstrate how different ensemble strategies perform on the same classification problem.

5.4 Model Training and Implementation

The implementation was carried out using Python and machine learning libraries, including Pandas, Scikit-learn, and XGBoost.

The main implementation steps were:

1. **Dataset loading:** The dataset was loaded into a Pandas DataFrame.
2. **Data cleaning:** Missing values and duplicate records were checked, and duplicate records were removed.
3. **Feature encoding:** Categorical values were converted into numerical values using binary mapping.
4. **Feature-target separation:** The 16 input features were separated from the `class` target variable.
5. **Train-test splitting:** The dataset was divided into training and testing sets using an 80:20 stratified split.
6. **Random Forest training:** A Random Forest classifier with 100 trees was fitted to the training data.
7. **XGBoost training:** An XGBoost classifier with 100 estimators, maximum depth 3, and learning rate 0.1 was fitted to the same training data.
8. **Prediction:** Both models generated predictions for the test data.
9. **Evaluation:** The predictions were assessed using accuracy, precision, recall, F1-score, and confusion matrices.
10. **Cross-validation:** Five-fold stratified cross-validation was used to evaluate performance across multiple data splits.
11. **Model saving:** The trained models and feature names were saved for use in the web application.

5.5 Performance Evaluation Metrics

The models were evaluated using several classification metrics to obtain a broader understanding of their performance.

Accuracy

Accuracy measures the proportion of correctly classified records among all test records.

Accuracy = (TP + TN) / (TP + TN + FP + FN)

Precision

Precision measures the proportion of predicted Positive cases that were actually Positive.

Precision = TP / (TP + FP)

Recall

Recall measures the proportion of actual Positive cases correctly identified by the model.

Recall = TP / (TP + FN)

F1-Score

The F1-score is the harmonic mean of precision and recall.

F1-Score = 2 × (Precision × Recall) / (Precision + Recall)

Confusion Matrix

A confusion matrix summarizes the model's correct and incorrect predictions using four values:

- True Positive (TP): Actual Positive records correctly predicted as Positive.
- True Negative (TN): Actual Negative records correctly predicted as Negative.
- False Positive (FP): Actual Negative records incorrectly predicted as Positive.
- False Negative (FN): Actual Positive records incorrectly predicted as Negative.

Using these metrics provides more information than accuracy alone, particularly because false negatives are important to consider in a preliminary health-screening application.

5.6 Experimental Results and Model Comparison

After duplicate removal and preprocessing, both classifiers were evaluated on the same test set containing 51 records.

Evaluation Metric	Random Forest	XG Boost
Accuracy	92.16%	88.24%
Precision	100.00%	100.00%
Recall	88.57%	82.86%
F1-Score	93.94%	90.63%
True Negatives	16	16

False Positives	0	0
False Negatives	4	6
True Positives	31	29

Random Forest achieved higher accuracy, recall, and F1-score than XGBoost on this test split. Both models achieved 100% precision and correctly classified all 16 Negative test records. Random Forest identified 31 Positive records, whereas XGBoost identified 29.

Five-fold cross-validation produced mean accuracies of 93.64% for Random Forest and 91.28% for XGBoost. These results also favored Random Forest on the selected dataset. However, the dataset is relatively small after duplicate removal, so the results should be interpreted cautiously and should not be considered evidence of clinical diagnostic performance.

5.7 Web Application Integration

The trained models were integrated into a web application using Flask, a lightweight Python web framework. The interface allows users to enter patient information, including age, gender, and symptoms.

The application follows these steps:

1. The user enters patient information through the web interface.
2. The frontend sends the input data to the Flask backend.
3. The backend converts categorical inputs into the numerical format used during training.
4. The input features are arranged in the correct order expected by the models.
5. The saved Random Forest and XGBoost models generate predictions.
6. The application displays the predictions from both classifiers to the user.

The trained models are saved so that the application can load them without retraining whenever a prediction is requested. This separates model development from model usage and makes the application easier to operate.

The web application demonstrates the practical integration of data preprocessing, machine learning classification, and frontend-backend communication. Its predictions are intended for educational and preliminary screening-support purposes and must not be treated as medical diagnoses.

CONCLUSION

The project titled **"Early-Stage Diabetes Risk Prediction Using Random Forest and XGBoost Classifiers"** demonstrates the application of machine learning techniques to classify patient records using demographic information and health-related symptoms.

The UCI Early Stage Diabetes Risk Prediction Dataset was analyzed through Exploratory Data Analysis to understand its structure, missing values, duplicate records, class distribution, age distribution, gender distribution, symptom patterns, and feature correlations. The original dataset contained 520 records, with 269 duplicate records identified and removed. After cleaning, 251 unique records remained. Categorical variables were converted into numerical representations, and the processed dataset was divided into training and testing sets using an 80:20 stratified split.

Two ensemble learning algorithms, Random Forest and XGBoost, were implemented and evaluated using accuracy, precision, recall, F1-score, confusion matrices, and five-fold cross-validation. On the selected test split, Random Forest achieved 92.16% accuracy and a 93.94% F1-score, while XGBoost achieved 88.24% accuracy and a 90.63% F1-score. Random Forest performed better on the reported test metrics and achieved a higher mean cross-validation accuracy.

The trained models were subsequently integrated into a Flask-based web application. The application enables users to enter patient information and obtain predictions from both classifiers through a simple interface. This integration demonstrates how machine learning models can be incorporated into an interactive software application.

Despite the encouraging results, the project has limitations, including the relatively small number of unique records, the restricted set of input features, and the possibility of incorrect predictions. The model performance obtained on this dataset does not guarantee similar results for other populations or real-world clinical settings.

Future improvements may include collecting larger and more diverse datasets, incorporating additional clinical parameters, conducting more extensive hyperparameter optimization, evaluating the models on independent datasets, and improving prediction interpretability.

In conclusion, this project provides practical experience in data analysis, data preprocessing, ensemble learning, model evaluation, and web application development. The developed system serves as an educational demonstration of preliminary diabetes risk prediction and is not a substitute for professional medical evaluation, laboratory testing, or clinical diagnosis.